

ACADEMIC TASK

Title:

IoT Device Dashboard for Smart Homes

Course: INT222 / Semester: 4 / Section: 3M042



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INT222 Project Report Submission

Course Code: INT222

Submission Details

Project Name	IoT Smart Homes
Project Title	IoT Device Dashboard for Smart Homes
Team Name	Harshit Kumar Nishad, Sahil Kumar, Himanka Saikia
Registration Number	12404530, 12407055, 12404571
GitHub Link	https://github.com/Harshxt-01/iot-smart-home
Deployed Link	https://iot-smart-home-ahl8.onrender.com/
Video Presentation Link	https://www.instagram.com/reel/DXmW2ihDMu1/?igsh=eHh0ZjJlbXppb2Ux

Project Overview

IoT Smart Home Dashboard is a full-stack web-based application designed to simulate a modern home automation system.

The project allows users to remotely monitor and control smart home devices such as lights, fans, and appliances through an interactive dashboard.

It combines a user-friendly frontend interface with a robust backend system to manage device states, user authentication, and real-time interactions.

The system demonstrates how IoT concepts can be integrated with web technologies to build a scalable and efficient smart home solution.

Key Features

- User authentication system (Login/Register) for secure access
- Interactive dashboard displaying all smart devices
- Device control system (Turn ON/OFF functionality)
- Real-time status updates (Online/Offline indication)
- Backend API integration for handling device operations
- MongoDB database for storing users and device data
- Cloud deployment for 24/7 accessibility
- Basic forgot password interface (UI-based)
- Scalable architecture for future IoT hardware integration

Technology Stack

- Frontend: HTML, CSS, Bootstrap
- Backend: Node.js, Express.js
- Database: MongoDB with Mongoose
- Language: TypeScript
- Deployment: Render (Cloud Hosting)
- Supporting Tools:
Nodemon (development)
dotenv (environment variables)
Morgan (logging)
CORS (cross-origin requests handling)